

YZV202E Optimization for Data Science Project Proposal Multi-Criteria Location Optimization for Data Center Placement in Türkiye

Osmancan Sari
AI and Data Engineering
Faculty of Computer and Informatics
sario20@itu.edu.tr
150230728

Yusuf Oğuz
AI and Data Engineering
Faculty of Computer and Informatics
oguzyu22@itu.edu.tr
150220322

Efe Karan Hacımustafaoğlu
AI and Data Engineering
Faculty of Computer and Informatics
hacimustafaoğlu24@itu.edu.tr
150240313

I. PROJECT DESCRIPTION

In this project, we are addressing the problem where to optimally place new data centers among Türkiye's 81 provinces. We aim to develop a model that identifies the best K cities by evaluating multiple metrics like renewable energy potential, cooling costs, seismic risk and distance between centers. By feeding our model with official open data from AFAD, TÜİK, MGM NASA, OpenStreetMap and KGM, we want to provide a realistic approach to planning procedures of large-scale tech investments.

II. PROBLEM DEFINITION

Working on this project, we frame the data center placement problem as a multi-criteria facility location task. Out of Türkiye's 81 provinces, our goal is to pinpoint the optimal cities for investment. We achieve this by evaluating an overall suitability score for each province, then selecting them according to budget or amount rules while respecting the physical and economic constraints of the real world.

A. Suitability Score

To do selection and make data visualization possible, we start by calculating the suitability score x_i for each province:

$$x_i = (w_1 F_i + w_2 P_i + w_3 L_i + w_4 E_i + w_5 C_i) \quad (1)$$

- i indexes the 81 provinces, following the Turkish license plate numbering convention (1–81).
- $\sum_{j=1}^5 w_j = 1, w_j \geq 0$
- F_i : Normalized land availability score.
- P_i : Normalized seismic safety score (inverse PGA values).
- L_i : Normalized labor density score (TÜİK data).
- E_i : Normalized renewable energy potential score, combining solar (NASA POWER GHI) and wind (TÜREB installed capacity) contributions.
- C_i : Normalized cooling efficiency (inverse average annual temperatures).

- w_j : Weight coefficients assigned to each criterion.

B. Objective Function

After the evaluation is done, the selection phase is initialized. Let y_i be a binary selection based on the province i where if province $y_i = 1$ if province i selected, $y_i = 0$ otherwise. The selection will have two modes.

- 1) **Budget Mode:** The model maximizes the suitability scores of selected provinces without exceeding the user specified budget:

$$\text{Maximize } Z_1 = \sum_{i=1}^{81} x_i y_i \quad \text{subject to } \sum_{i=1}^{81} c_i y_i \leq B \quad (2)$$

where c_i is estimated province-specific cost.

- 2) **Count Mode:** The user specifies the number of provinces K to be selected. Then the model maximizes the suitability score:

$$\text{Maximize } Z_2 = \sum_{i=1}^{81} x_i y_i \quad \text{subject to } \sum_{i=1}^{81} y_i = K \quad (3)$$

III. DATASET

- 1) **Disaster and Emergency Management Presidency (AFAD):** Published data from the website will be used to retrieve earthquake risk data. The Excel file that we obtained from this site include different PGA values for different scenarios for fixed spaced coordinates. “%10 in 50 year” scenarios will be used as standard approach. Each coordinate will be matched to their province, and detect average earthquake risk for each city (AFAD, 2026).
- 2) **Turkish Statistical Institute (TÜİK):** This statistical hub of Türkiye will be used to retrieve labor data. For the labor density, multiplication of two different datasets will be used. First one is employment rate, second one is total working age(+15) population of each city. For the

budget constraint, we will pull average income values by region (TÜİK, 2026).

- 3) **Turkish State Meteorological Service (MGM):** The website will be used to retrieve annual average temperature data. Because of unavailability of a structured data(for example Excel file), these data will be obtained by web scraping (MGM, 2026).
- 4) **NASA POWER API:** This api will be used to retrieve annual average GHI data. Because GHI values represent the average amount of sunlight falling on the region, this data will be used as the solar component of the renewable energy potential criterion (NASA POWER, 2026).
- 5) **Turkish Wind Energy Association (TÜREB):** The biannual Turkish Wind Energy Statistics Report provides province-level installed wind power capacity (in MWm) for operational wind power plants across Türkiye, derived directly from turbine manufacturer data. The publicly available July 2022 full report will be used as the wind component of the renewable energy potential score (TÜREB, 2022).
- 6) **OpenStreetMap:** This source will be used to retrieve coordinates for each city center. This data will be accessed via the osmnx Python library (OpenStreetMap, 2026).
- 7) **General Directorate of Highways (KGM):** Distance for province to province data from the General Directorate of Highways (KGM) will be used to pull distance between any two provinces (KGM, 2026).
- 8) **Shuttle Radar Topography Mission (SRTM):** This is another NASA sourced dataset. SRTM will be used to pull the elevation data (NASA Earthdata, 2026).

IV. METHODOLOGY

This section explains the numeric handling of each score and completes the linear programming models by explaining the constraints. Each of the 81 provinces is represented by a single candidate point: the official city center coordinates retrieved from OpenStreetMap. All distance computations (province-to-province road network distance, distance to the southern border) and area-availability calculations are anchored to this point, providing a single, reproducible representative location per province.

A. Calculating the Suitability Score

- 1) **Land Availability:** Since Türkiye's terrain varies significantly in elevation, not all provinces offer sufficient flat land for large-scale infrastructure. Using SRTM elevation data, we compute the slope s at each grid point within a province. Let A_i be the total area (in km^2) of province i with slope below a buildable threshold θ :

$$A_i = |\{p \in \text{province}_i \mid s_p < \theta\}| \cdot r^2 \quad (4)$$

where r is the spatial resolution of the SRTM grid in kilometers. We use the SRTM-3 (3 arc-second, ≈ 90 m) global product, giving $r \approx 0.09$ km, which provides

sufficient detail for province-level slope aggregation while keeping the computation tractable across all 81 provinces. Standard Min-Max normalization is then applied:

$$F_i = \frac{A_i - A_{\min}}{A_{\max} - A_{\min}} \quad (5)$$

- 2) **Seismic Safety:** Our collected data consists of Peak Ground Acceleration (PGA) values, which increase with the increased risk. Because our objective function aims to maximize the suitability of a province, we need to inverse seismic risk to seismic safety. Since PGA values are well-established in the seismic hazard literature to follow an approximately log-normal distribution (Cornell, 1968; Baker, 2013), logarithmic scaling yields more accurate normalization, therefore we map the minimum and the maximum value to 0 and 1 logarithmically. Let PGA_i be the collected raw PGA value for province i , then the safety metric P_i is defined as:

$$P_i = 1 - \frac{\ln(PGA_i) - \ln(PGA_{\min})}{\ln(PGA_{\max}) - \ln(PGA_{\min})} \quad (6)$$

where $PGA_{\max}$ and $PGA_{\min}$ represent maximum and minimum values observed from our dataset for 81 provinces, respectively.

- 3) **Labor Density:** Since we don't have the labor force metric directly from our sources, we derive it from two different values; average employment rate and total population of working age. Let N_i and η_i be total population of working age and average employment rate for province i , respectively:

$$L_{\text{raw},i} = N_i \eta_i \quad (7)$$

where $L_{\text{raw},i}$ is the raw labor force of province i . Since labor suitability doesn't scale linearly with the raw labor force in larger populations we scale it down logarithmically:

$$L_{\text{adj},i} = \ln(L_{\text{raw},i}) \quad (8)$$

Then we apply standard Min-Max normalization where we map the minimum to 0 and the maximum to 1 and scale the other values accordingly. Let $L_{\max}$ and $L_{\min}$ be the maximum and minimum values collected from the logarithmically adjusted values, then the labor density score L_i is:

$$L_i = \frac{L_{\text{adj},i} - L_{\min}}{L_{\max} - L_{\min}} \quad (9)$$

- 4) **Renewable Energy Potential Score:** The renewable energy potential is composed of two sources: solar irradiance (GHI) and operational wind capacity. Let S_i be the average annual GHI value for province i , and $W_{\text{raw},i}$ be the installed wind power capacity (in MWm) of operational wind power plants in province i as reported by TÜREB. Since the provincial wind capacity distribution is heavily skewed (a few provinces such as İzmir and Balıkesir host the majority of installed capacity, while many provinces have zero), we apply a

logarithmic adjustment with a +1 shift to handle zero values:

$$W_{adj,i} = \ln(W_{raw,i} + 1) \quad (10)$$

Each component is then normalized using standard Min-Max normalization:

$$\tilde{S}_i = \frac{S_i - S_{min}}{S_{max} - S_{min}}, \quad \tilde{W}_i = \frac{W_{adj,i} - W_{adj,min}}{W_{adj,max} - W_{adj,min}} \quad (11)$$

The combined renewable energy potential score E_i is calculated as a linear combination with internal weights $\alpha + \beta = 1$:

$$E_i = \alpha \tilde{S}_i + \beta \tilde{W}_i \quad (12)$$

Since solar potential is uniformly available across all 81 provinces while wind is highly localized to specific geographic corridors (Aegean, Marmara), the default internal weights favor solar slightly: $\alpha = 0.6$, $\beta = 0.4$. Note that using *installed* wind capacity rather than raw wind speed is a deliberate choice: a high installed capacity simultaneously signals favorable wind conditions, existing grid integration, and proven investor interest, all of which are directly relevant to data center siting.

- 5) **Cooling Efficiency:** Cooling Efficiency score is handled similarly to the Renewable Energy Potential Score, using standard Min-Max normalization but inverting the values as lower average temperature means less cooling cost, therefore more efficient cooling. Let T_i be the average temperature for a province i :

$$C_i = 1 - \frac{T_i - T_{min}}{T_{max} - T_{min}} \quad (13)$$

where T_{max} and T_{min} represent maximum and minimum values from our data, respectively.

B. Preprocessing

To incorporate into the solver, we need to calculate some constraint values before the solver.

- 1) **Minimum Distance Constraint:** To be ready for a geographical disaster, the two chosen provinces i and j must be at least D kilometers apart in road network distance-calculated from the official city centers-as the varying elevation mentioned in land availability section makes straight line calculation inaccurate. Road network distance values are pulled from KGM. To preserve the linearity of the model we pre-compute all the *forbidden pairs* and map them inside a function:

$$\mathcal{F} = \{(i, j) \mid i < j, \text{dist}(i, j) < D\} \quad (14)$$

where $\text{dist}(i, j)$ is the distance between provinces i and j .

- 2) **National Security Constraint:** Türkiye's real national security concerns lie on its southern border, and Türkiye's Syria and Iraq borders run roughly along the same 37th parallel (Wikipedia, 2026), making a single reference latitude $\phi_{border} = 37 \times \frac{\pi}{180}$ a reasonable geometric approximation for the entire southern border.

The orthogonal distance d_i from the official center of province i to the border is calculated as:

$$d_i = R \cdot |\phi_i - \phi_{border}| \quad (15)$$

where ϕ_i is the latitude of province i , pulled from OpenStreetMap data. Provinces that do not meet the minimum safe distance D_{safe} are eliminated from the selection:

$$y_i = 0 \quad \forall i \in \{i \mid d_i < D_{safe}\} \quad (16)$$

The 37th-parallel reference is a deliberate simplification rather than a strict geographic match. The actual Syria-Türkiye and Iraq-Türkiye borders deviate from $\phi = 37^\circ$ by at most ~ 50 km along the entire southern frontier, with the largest deviations occurring in the easternmost provinces (Hakkari, Şırnak). Since D_{safe} is itself a tunable parameter exposed to the user via a slider, any conservative estimation can be compensated by increasing D_{safe} accordingly. A polyline-based exact-distance computation against the full border geometry was considered but not adopted, as it would not change the set of eliminated provinces under realistic D_{safe} values (typically 100–200 km) while introducing additional preprocessing complexity.

- 3) **Cost Estimation:** To satisfy budget constraints defined in Mode 2 (2), we estimate c_i , the total cost of establishing and operating a data center in province i . We derive c_i by combining a constant base cost, regional personnel expenses, and national energy costs. Let C_{base} be the base cost to build a data center. Let I_i be the average annual income in province i , and P_{exp} be the expected number of personnel required to operate the data center. The annual personnel cost is calculated strictly as the product of the required headcount and the local average income. With E_{annual} as the estimated annual energy consumption of the facility (in kWh) and U_{elec} as the declared industrial electricity cost per unit in Türkiye (EPDK, 2026), the province-specific cost c_i is calculated as:

$$c_i = C_{base} + (P_{exp} \cdot I_i) + (E_{annual} \cdot U_{elec}) \quad (17)$$

where the regional variance is established by I_i , while energy and base cost provide the baseline.

C. Common Constraints

Both of the modes will also be subject to the same common constraints:

- 1) **Minimum Distance Constraint:** Since the impossible pairs were calculated and mapped before we just subject the model to:

$$y_i + y_j \leq 1 \quad \forall (i, j) \in \mathcal{F} \quad (18)$$

- 2) **Minimum Seismic Safety Constraint:** To protect the data center network, the total seismic safety score of the provinces chosen must be above a certain level R_{min} :

$$\sum_{i=1}^{81} P_i y_i \geq R_{min} \quad (19)$$

D. Weights, Constants and Variables, Solver and Data Visualization

- 1) **Weights:** The weights coefficients $w_1 \dots w_5$ will be determined using the Analytic Hierarchy Process (AHP), where each criterion is compared pairwise against every other criterion using a 1–9 importance scale. The algebraic way to implement this process is to create a matrix M where entry m_{kl} represents how much more important criterion k is over criterion l :

$$M = \begin{pmatrix} 1 & m_{12} & m_{13} & m_{14} & m_{15} \\ 1/m_{12} & 1 & m_{23} & m_{24} & m_{25} \\ 1/m_{13} & 1/m_{23} & 1 & m_{34} & m_{35} \\ 1/m_{14} & 1/m_{24} & 1/m_{34} & 1 & m_{45} \\ 1/m_{15} & 1/m_{25} & 1/m_{35} & 1/m_{45} & 1 \end{pmatrix} \quad (20)$$

For instance, if seismic safety is judged *strongly more important* than labor density on Saaty’s 1–9 scale, then $m_{23} = 5$, and consequently $m_{32} = 1/5$. The principal eigenvector of this matrix represents the weights according to our hierarchy, which is then normalized so that the components sum to 1, consistent with the constraint $\sum_{j=1}^5 w_j = 1$. Across all default presets, the labor density criterion is assigned a relatively low weight, reflecting the fact that skilled IT personnel can relocate and labor availability should not dominate the placement decision. Since the end goal is to create a customizable tool to determine the best provinces according to different needs, the final version will include sliders to change the weights as well as different presets of weight profiling such as equal-weight, seismic-priority, cooling-priority and renewable-priority. In addition to the weight sliders, the interface will also provide sliders for the selection target (K in Count Mode or B in Budget Mode) and the key constraint parameters D , D_{safe} , R_{min} , and θ , so that final experiments can test multiple weight and constraint scenarios.

- 2) **Constants and Variables:** As per our main goal customizability, constant values we assumed in many calculations will be variables left to user discretion. However, sensible defaults will be assigned after the foundation to use them is laid. This is the list of all constants we assumed and their equation references:
- θ : Maximum slope for buildable threshold. (4)
 - α, β : Coefficient of internal weights for solar and wind availability, respectively. (12)
 - D : Minimum road network distance between any two provinces. (14)
 - D_{safe} : Minimum straight line distance from southern border. (16)

- $C_{base}, P_{exp}, E_{annual}$: Estimated values for base cost to build a mid-sized data center, number of personnel, annual energy consumption respectively. (17)
- U_{elec} : Declared industrial energy cost per unit. (17)
- R_{min} : Minimum seismic safety score threshold. (19)

The modes of the objective function require user input:

- B : Maximum budget. (2)
- K : Number of provinces to be selected. (3)

- 3) **Solver:** Since both selection modes are defined as Binary Integer Programs we will implement Branch and Bound algorithm where the problem is broken into smaller subproblems and Linear Programming relaxations are used to determine whether this subproblem includes an optimal solution, using PuLP modeling library (Peschiera et al., 2026). Although Branch and Bound has worst-case exponential complexity in the number of binary variables, the model contains only 81 binary variables and a small constraint set, which keeps the branching tree tractable; the PuLP solver handles instances of this size within seconds.
- 4) **Data Visualization:** The project will demonstrate the result by utilizing a choropleth map of Türkiye where each province is colored according to its suitability score. As mentioned, sliders and other interactive elements will be added to allow the user to change criteria according to their needs. Province borders for the choropleth map will be pulled from GADM shapefiles (GADM, 2026) using the GeoPandas library (GeoPandas developers, 2026) and will be colorized using matplotlib (Matplotlib development team, 2026) library.

V. LIMITATIONS

- Project only optimizes provinces on their average values and using their city center coordinates.
- As area availability is introduced as a suitability factor, program will already prefer Central Anatolia Region provinces as they have more large flat areas compared to other regions. Because of that National Security Constraints only considers southern border, as using the closest border would push the selections further to Central Anatolia Region.
- Youth unemployment and Eurostat’s NEET (Young persons Neither in Employment nor in Education and Training) indicator were considered but not incorporated into the labor density score, as the direct applicability of the NEET indicator to the Turkish labor market is limited for several reasons. First, NEET is an aggregate signal that cannot be straightforwardly mapped to skilled-IT availability: a high NEET rate may equally reflect an eager, untapped workforce or a structurally underqualified labor pool that does not match data center personnel requirements, making the indicator simultaneously penalizing and rewarding for the optimization. Second, the demographic and educational composition behind NEET

differs significantly between Türkiye and the EU member states for which Eurostat’s methodology was originally designed, so its scoring direction (higher vs. lower being preferable) is not unambiguous in this context. Third, since skilled IT personnel are highly mobile and labor density is intentionally assigned a low weight in our AHP setup, introducing an additional weakly-signaling NEET term would add noise without meaningfully improving the discrimination between provinces. For these reasons, the labor score is restricted to the logarithmically scaled working-age workforce, and NEET-driven extensions are left as future work once a Türkiye-specific calibration of the indicator becomes available.

- Nuclear capacity was considered as a third component of the renewable energy potential score but ultimately excluded from the model. As of the time of writing, Türkiye has only one licensed and partially operational nuclear site (Akkuyu in Mersin, 4,800 MW planned) and two additional projects in early stages (Sinop with site approval granted in 2024, and the Trakya/Kıyıköy project in Kırklareli with only site selection finalized in 2023). Incorporating nuclear capacity would therefore reduce to a binary or near-binary signal that rewards a single province (Mersin) while leaving 78–80 provinces at zero, producing a sparse and unstable feature rather than a continuous criterion. We adopt the principle of including a criterion only when a complete, province-level dataset is available; nuclear can be revisited once additional sites become operational.
- The wind component of the renewable energy potential score is based on the publicly available TÜREB Turkish Wind Energy Statistics Report dated July 2022, as more recent full reports are restricted to TÜREB members. Provincial rankings of installed wind capacity have remained stable, but absolute values are conservative for high-growth provinces where significant capacity may have been added since 2022. Updating to a newer release once it becomes publicly accessible is a straightforward future revision and does not affect the model’s structure.

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